7 cognition: thinking, intelligence, and language

Think about how you interact with the world around you. How often do you simply respond, without knowing how or why you do the things you do, say, or think? How much of your conscious experience involves effortful, mindful attention, and decision making? These two types of thinking, sometimes referred to as System 1 and System 2, characterize much of how we think and process information (Kahneman, 2011; Stanovich & West, 2000). System 1, which involves making quick decisions and using cognitive shortcuts, is guided by our innate abilities and personal experiences. System 2, which is relatively slow, analytical, and rule-based, is dependent more on our formal educational experiences. Overall, our thinking has to be governed by the interplay between the two.

Do you tend to rely more on instinctual (System 1) or deliberate (System 2) thought processes? How do your thought processes and decision making strategies vary depending on the situation?

Watch the Video at MyPsychLab.com

Why study the nature of thought?

To fully understand how we do any of the things we do (such as learning, remembering, and behaving), we need to understand how we think. How do we organize our thoughts? How do we communicate those thoughts to others? What do we mean by intelligence? Why are some people able to learn so much faster than others?

learning objectives

7.1 How are mental images and concepts involved in the process of thinking?

7.2 What are the methods people use to solve problems and make decisions?

7.3 Why does problem solving sometimes fail, and what is meant by creative thinking?

7.4 How do psychologists define intelligence, and how do various theories of intelligence differ?

7.5 How is intelligence measured, how are intelligence tests constructed, and what role do these tests play in neuropsychology?

7.6 What is intellectual disability and what are its causes?

7.7 What defines giftedness, and how are giftedness and emotional intelligence related to success in life?

7.8 What is the influence of heredity and environment on the development of intelligence?

7.9 How is language defined, and what are its different elements and structure?

7.10 Does language influence the way people think, and are animals capable of learning language?

7.11 What are some ways to improve thinking?

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How People Think

What does it mean to think? People are thinking all the time and talking about thinking as well: "What do you think?" "Let me think about that." "I don't think so." So, what does it mean to think? **Thinking**, or **cognition** (from a Latin word meaning "to know"), can be defined as mental activity that goes on in the brain when a person is processing information—organizing it, understanding it, and communicating it to others. Thinking includes memory, but it is much more. When people think, they are not only aware of the information in the brain but also are making decisions about it, comparing it to other information, and using it to solve problems.

Thinking also includes more than just a kind of verbal "stream of consciousness." When people think, they often have images as well as words in their minds.

MENTAL IMAGERY

How are mental images and concepts involved in the process of thinking?

As stated in Chapter Six, short-term memories are encoded in the form of sounds and also as visual images, forming a mental picture of the world. Thus, **mental images** (representations that stand in for objects or events and have a picturelike quality) are one of several tools used in the thought process.

Here's an interesting demonstration of the use of mental images. Get several people together and ask them to tell you *as fast as they can* how many windows are in the place where they live. Usually you'll find that the first people to shout out an answer have fewer windows in their houses than the ones who take longer to respond. You'll also notice that most of them look up, as if looking at some image that only they can see. If asked, they'll say that to determine the number of windows, they pictured where they live and simply counted windows as they "walked through" the image they created in their mind.

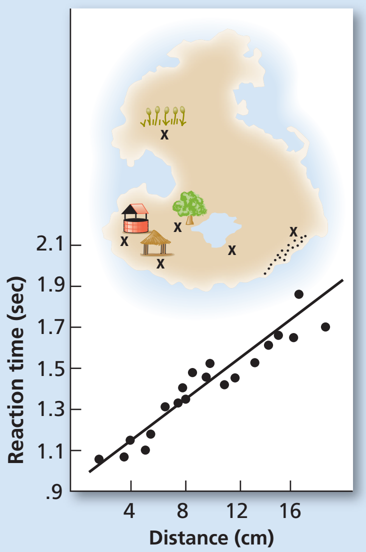


Figure 7.1 Kosslyn's Fictional Island

In Kosslyn's 1978 study, participants were asked to push a button when they had imagined themselves moving from one place on the island to another. As the graph below the picture shows, participants took longer times to complete the task when the locations on the image were farther apart.
Source: Kosslyn et al. (1978).

So more windows means more time to count them in your head? I guess mentally "walking" through a bigger house in your head would take longer than "walking" through a smaller one.

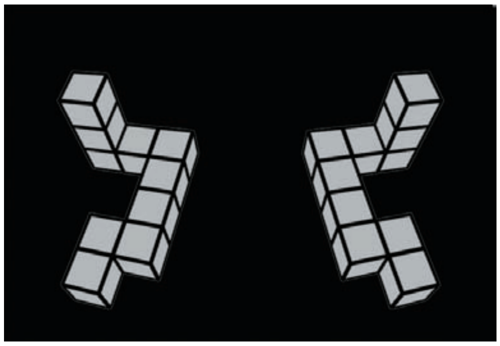
That's what researchers think, too. They have found that it does take longer to view a mental image that is larger or covers more distance than a smaller, more compact one (Kosslyn et al., 2001; Ochsner & Kosslyn, 1994). In one study (Kosslyn et al., 1978), participants were asked to look at a map of an imaginary island (see **Figure 7.1**). On this map were several landmarks, such as a hut, a lake, and a grassy area. After viewing the map and memorizing it, participants were asked to imagine a specific place on the island, such as the hut, and then to "look" for another place, like the lake. When they mentally "reached" the second place, they pushed a button that recorded reaction time. The greater the physical distance on the map between the two locations, the longer it took participants to scan the image for the second location. The participants were apparently looking at their mental image and scanning it just as if it were a real, physical map.

People are even able to mentally rotate, or turn, images (Shepherd & Metzler, 1971). Kosslyn (1983) asked participants questions such as the following: "Do frogs have lips and a stubby tail?" He found that most participants reported visualizing a frog, starting with the face ("no lips"), then mentally rotating the image so it was facing away from them, and then "zooming in" to look for the stubby tail ("yes, there it is"). A very important aspect of the research on mental rotation is that we tend to engage *mental* images in our mind much like we engage or interact with *physical* objects. When we rotate an object in our minds (or in other ways interact with or manipulate mental images), it is not instantaneous—it takes time, just as it would if we were rotating a physical object with our hands. To see how well you are able to mentally rotate images, try the *Mental Rotation* experiment.

Simulation

Mental Rotation

In this experiment, we will address your ability to mentally rotate objects in your mind. You will be presented with two objects, and asked to determine if the objects are the same except for their orientation.



Go to the Experiment ▶

Simulate the **Experiment**, *Mental Rotation*, on **MyPsychLab**

In the brain, creating a mental image is almost the opposite of seeing an actual image. With an actual image, the information goes from the eyes to the visual cortex of the occipital lobe and is processed, or interpreted, by other areas of the cortex that compare the new information to information already in memory. LINK to Learning Objective 2.9. In creating a mental image, areas of the cortex associated with stored knowledge send information to the visual cortex, where the image is perceived in the "mind's eye" (Kosslyn et al., 1993; Sparing et al., 2002). PET scans show areas of the visual cortex being activated during the process of forming an image, providing evidence for the role of the visual cortex in mental imagery (Kosslyn et al., 1993, 1999, 2001). **Watch** the **Video**, *Special Topics: Mental Imagery: In the Mind's Eye,* at **MyPsychLab**

Through the use of functional magnetic resonance imagery (fMRI), researchers have been able to see the overlap that occurs in brain areas activated during visual mental imagery tasks as compared to actual tasks involving visual perception (Ganis et al., 2004). During both types of tasks, activity was present in the frontal cortex (cognitive control), temporal lobes (memory), parietal lobes (attention and spatial memory), and occipital lobes (visual processing). However, the amount of activity in these areas differed between the two types of tasks. For example, activity in the visual cortex was stronger during perception than in imagery, suggesting sensory input activates this area more strongly than memory input. And an important finding overall, those areas activated during visual imagery were a subset of those activated during visual

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perception, with the greatest similarity in the frontal and parietal regions rather than the temporal and occipital regions. What does this mean? Simply that there is commonality between the processes of visual imagery and visual perception but it *is not* a complete overlap, and, as the authors point out, the greater overlap *was not* in the temporal and occipital regions (memory and vision functions) that might be assumed to be the most likely areas of overlap given the visual nature of the tasks (Ganis et al., 2004).

CONCEPTS AND PROTOTYPES

Images are not the only way we think, are they?

Mental images are only one form of mental representation. Another aspect of thought processes is the use of concepts. **Concepts** are ideas that represent a class or category of objects, events, or activities. People use concepts to think about objects or events without having to think about all the specific examples of the category. For example, a person can think about "fruit" without thinking about every kind of fruit there is in the world, which would take far more effort and time. This ability to think in terms of concepts allows us to communicate with each other: If I mention a bird to you, you know what I am referring to, even if we aren't actually thinking of the same *type* of bird.

Concepts not only contain the important features of the objects or events people want to think about, but also they allow the identification of new objects and events that may fit the concept. For example, dogs come in all shapes, sizes, colors, and lengths of fur. Yet most people have no trouble recognizing dogs as dogs, even though they may never before have seen that particular breed of dog. Friends of the author have a dog called a briard, which is a kind of sheepdog. In spite of the fact that this dog is easily the size of a small pony, the author had no trouble recognizing it as a dog, albeit a huge and extremely shaggy one.

Concepts can have very strict definitions, such as the concept of a square as a shape with four equal sides. Concepts defined by specific rules or features are called *formal concepts* and are quite rigid. To be a square, for example, an object must be a two-dimensional figure with four equal sides and four angles adding up to 360 degrees. Mathematics is full of formal concepts. For example, in geometry there are triangles, squares, rectangles,

Both of these animals are dogs. They both have fur, four legs, a tail—but the similarities end there. With so many variations in the animals we call "dogs," what is the prototype for "dog"?

polygons, and lines. In psychology, there are double-blind experiments, sleep stages, and conditioned stimuli, to name a few. Each of these concepts must fit very specific features to be considered true examples.

A duck-billed platypus is classified as a mammal yet shares features with birds, such as webbed feet and a bill, and it also lays eggs. The platypus is an example of a "fuzzy" natural concept. Courtesy of Dave Watts, Nature Picture Library.

But what about things that don't easily fit the rules or features? What if a thing has some, but not all, features of a concept?

People are surrounded by objects, events, and activities that are not as clearly defined as formal concepts. What is a vehicle? Cars and trucks leap immediately to mind, but what about a bobsled, or a raft? Those last two objects aren't quite as easy to classify as vehicles immediately, but they fit some of the rules for "vehicle." These are examples of *natural concepts*, concepts people form not as a result of a strict set of rules, but rather as the result of experiences with these concepts in the real world (Ahn, 1998; Barton & Komatsu, 1989; Rosch, 1973). Formal concepts are well defined, but natural concepts are "fuzzy" (Hampton, 1998). Natural concepts are important in helping people understand their surroundings in a less structured manner than school-taught formal concepts, and they form the basis for interpreting those surroundings and the events that may occur in everyday life.

When someone says "fruit," what's the first image that comes to mind? More than likely, it's a specific kind of fruit like an apple, pear, or orange. It's less likely that someone's first impulse will be to say "guava" or "papaya," or even "banana," unless that person comes from a tropical area. In the United States, apples are a good example of a **prototype**, a concept that closely matches the defining characteristics of the concept (Mervis & Rosch, 1981; Rosch, 1977). Fruit is sweet, grows on trees, has seeds, and is usually round—all very applelike qualities. Coconuts are sweet and they also grow on trees, but many people in the Northern Hemisphere have never actually seen a coconut tree. They have more likely seen countless apple trees. So people who do have very different experiences with fruit, for instance, will have different prototypes, which are the most basic examples of concepts.

What about people who live in a tropical area? Would their prototype for fruit be different? And would people's prototypes vary in other cultures?

More than likely, prototypes develop according to the exposure a person has to objects in that category. So someone who grew up in an area where there are many coconut trees might think of coconuts as more prototypical than apples, whereas someone growing up in the northwestern United States would more likely see apples as a prototypical fruit (Aitchison, 1992). Culture also matters in the formation of prototypes. Research on concept prototypes across various cultures found greater differences and variations in prototypes between cultures that were dissimilar, such as Taiwan and America, than between cultures that are more similar, such as Hispanic Americans and non–Hispanic Americans living in Florida (Lin et al., 1990; Lin & Schwanenflugel, 1995; Schwanenflugel & Rey, 1986).

How do prototypes affect thinking? People tend to look at potential examples of a concept and compare them to the prototype to see how well they match—which is why it takes most people much longer to think about olives and tomatoes as fruit because they aren't sweet, one of the major characteristics of the prototype of fruit (Rosch & Mervis, 1975). As the video *The Basics: The Mind Is What the Brain Does* explains, we use a combination of cognitive processes including concepts, prototypes, and mental images to identify objects in our daily lives.

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Watch the **Video**, *The Basics: The Mind Is What the Brain Does,* at **MyPsychLab**

No matter what type, concepts are one of the ways people deal with all the information that bombards* their senses every day, allowing them to organize their perceptions of the world around them. This organization may take the form of *schemas*, mental generalizations about objects, places, events, and people (for example, one's schema for "library" would no doubt include books and bookshelves), or *scripts*, a kind of schema that involves a familiar sequence of activities (for example, "going to a movie" would include traveling there, getting the ticket, buying snacks, finding the right theater, etc.). Concepts not only help people think, but also they are an important tool in *problem solving*, a type of thinking that people engage in every day and in many different situations.

PROBLEM-SOLVING AND DECISION-MAKING STRATEGIES

What are the methods people use to solve problems and make decisions?

Problem solving is certainly a big part of any college student's life. Is there any one "best" way to go about solving a problem?

This child may try one piece after another until finding the piece that fits. This is an example of trial-and-error learning.

Think about it as you read on and solve the following: Put a coin in a bottle and then cork the opening. How can you get the coin out of the bottle without pulling out the cork or breaking the bottle? (For the solution, see the section on Insight.)

As stated earlier, images and concepts are mental tools that can be used to solve problems and make decisions. For the preceding problem, you are probably trying to create an image of the bottle with a coin in it. **Problem solving** occurs when a goal must be reached by thinking and behaving in certain ways. Problems range from figuring out how to cut a recipe in half to understanding complex mathematical proofs to deciding what to major in at college. Problem solving is one aspect of **decision making**, or identifying, evaluating, and choosing among several alternatives. There are several different ways in which people can think in order to solve problems. **Watch** the **Video**, *In the Real World: Changing Your Mind*, at **MyPsychLab**

TRIAL AND ERROR (MECHANICAL SOLUTIONS) One method is to use **trial and error**, also known as a **mechanical solution**. Trial and error refers to trying one solution after another

*bombards: attacks again and again.

until finding one that works. For example, if Shelana has forgotten the PIN for her online banking Web site, she can try one combination after another until she finds the one that works, if she has only a few such PINs that she normally uses. Mechanical solutions can also involve solving by *rote*, or a learned set of rules. This is how word problems were solved in grade school, for example. One type of rote solution is to use an algorithm.

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ALGORITHMS **Algorithms** are specific, step-by-step procedures for solving certain types of problems. Algorithms will always result in a correct solution, if there is a correct solution to be found, and you have enough time to find it. Mathematical formulas are algorithms. When librarians organize books on bookshelves, they also use an algorithm: Place books in alphabetical order within each category, for example. Many puzzles, like a Rubik's Cube®, have a set of steps that, if followed exactly, will always result in solving the puzzle. But algorithms aren't always practical to use. For example, if Shelana didn't have a clue what those four numbers might be, she *might* be able to figure out her forgotten PIN by trying *all possible combinations* of four digits, 0 through 9. She would eventually find the right four-digit combination—but it might take a very long while! Computers, however, can run searches like this one very quickly, so the systematic search algorithm is a useful part of some computer programs.

HEURISTICS Unfortunately, humans aren't as fast as computers and need some other way to narrow down the possible solutions to only a few. One way to do this is to use a heuristic. A **heuristic**, or "rule of thumb," is a simple rule that is intended to apply to many situations. Whereas an algorithm is very specific and will always lead to a solution, a heuristic is an educated guess based on prior experiences that helps narrow down the possible solutions for a problem. For example, if a student is typing a paper in a word-processing program and wants to know how to format the page, he or she could try to read an entire manual on the word-processing program. That would take a while. Instead, the student could use an Internet search engine or type "format" into the help feature's search program. Doing either action greatly reduces the amount of information the student will have to look at to get an answer. Using the help feature or clicking on the appropriate toolbar word will also work for similar problems. **Watch** the **Video**, *What's in It for Me?: Making Choices*, at **MyPsychLab**

Representativeness Heuristic Will using a rule of thumb always work, like algorithms do? Using a heuristic is faster than using an algorithm in many cases, but unlike algorithms, heuristics will *not* always lead to the correct solution. What you gain in speed is sometimes lost in accuracy. For example, a **representativeness heuristic** is used for categorizing objects and simply assumes that any object (or person) that shares characteristics with the members of a particular category is also a member of that category. This is a handy tool when it comes to classifying plants but doesn't work as well when applied to people. The representativeness heuristic can cause errors due to ignoring base rates, the actual probability of a given event. Are all people with dark skin from Africa? Does everyone with red hair also have a bad temper? Are all blue-eyed blondes from Sweden? See the point? The representativeness heuristic can be used—or misused—to create and sustain stereotypes (Kahneman & Tversky, 1973; Kahneman et al., 1982).

Availability Heuristic Another heuristic that can have undesired outcomes is the **availability heuristic**, which is based on our estimation of the frequency or likelihood of an event based on how easy it is to recall relevant information from memory or how easy it is for us to think of related examples (Tversky & Kahneman, 1973). Imagine, for example, that after you have already read this entire textbook (it could happen!) you are asked to estimate how many words in the book start with the letter *K* and how many have the letter *K* as the third letter in the word. Which place do you think is more frequent, the first letter or as the third letter? Next, what do you think the ratio of the more frequent placement is to the less frequent placement? What is easier to think of, words that begin with the letter *K* or words that have *K* as the third letter? Tversky & Kahneman (1973) asked this same question of 152 participants for five consonants (*K, N, L, R, V*) that appear more frequently in the third position as compared to the first in a typical text. Sixty-nine percent of the participants indicated

Smartphones and other portable devices provide tools for easy navigation. How might the use or overuse of these tools affect our ability to navigate when we do not have access to them?

that the first position was the more frequent placement and the median estimated ratio was 2:1 for the letter *K*—however, there are typically twice as many words with *K* as the third letter as compared to the first. Can you think of an example where you may have used the availability heuristic and it did not work in your favor?

Working Backward A useful heuristic that *does* work much of the time is to *work backward from the goal.* For example, if you want to know the shortest way to get to the new coffee shop in town, you already know the goal, which is finding the coffee shop. There are probably several ways to get there from your house, and some are shorter than others. Assuming you have the address of the store, for many the best way to determine the shortest route is to look up the location of the store on an Internet map, a GPS, or a smartphone, and compare the different routes by the means of travel (walking versus driving). People actually used to do this with a physical map and compare the routes manually! Think about it, does technology help or hinder some aspects of problem solving? What are, if any, the benefits to using technology for solving some problems as compared to actively engaging in problem solving as a mental challenge?

What if my problem is writing a term paper? Starting at the end isn't going to help me much!

One rule of thumb, or heuristic, involves breaking down a goal into subgoals. This woman is consulting the map to see which of several possible paths she needs to take to get to her goal destination.

Subgoals Sometimes it's better to break a goal down into *subgoals*, so that as each subgoal is achieved, the final solution is that much closer. Writing a term paper, for example, can seem overwhelming until it is broken down into steps: Choose a topic, research the topic, organize what has been gathered, write one section at a time, and so on, LINK to Learning Objective PIA.6. Other examples of heuristics include making diagrams to help organize the information concerning the problem or testing possible solutions to the problem one by one and eliminating those that do not work.

Sometimes I have to find answers to problems one step at a time, but in other cases the answer seems to just "pop" into my head all of a sudden. Why do some answers come so easily to mind?

INSIGHT When the solution to a problem seems to come suddenly to mind, it is called insight. Chapter Five contained a discussion of Köhler's (1925) work with Sultan the chimpanzee, which demonstrated that even some animals can solve problems by means of a sudden insight. LINK to Learning Objective 5.11. In humans, insight often takes the form of an "aha!" moment—the solution seems to come in a flash. A person may realize that this problem is similar to another one that he or she already knows how to solve or might see that an object can be used for a different purpose than its original one, like using a dime as a screwdriver.

Remember the problem of the bottle discussed earlier in this chapter? The task was to get the coin out of the bottle without removing the cork or breaking the bottle. The answer is simple: *Push the cork into the bottle and shake out the coin. Aha!*

Insight is not really a magical process, although it can seem like magic. What usually happens is that the mind simply reorganizes a problem, sometimes while the person is thinking about something else (Durso et al., 1994).

Here's a problem that can be solved with insight: Marsha and Marjorie were born on the same day of the same month of the same year to the same mother and the same father yet they are not twins. How is that possible? Think about it and then look for the answer in the section on Mental Sets.

In summary, thinking is a complex process involving the use of mental imagery and various types of concepts to organize the events of daily life. Problem solving is a special type of thinking that involves the use of many tools, such as trial-and-error thinking, algorithms, and heuristics, to solve different types of problems **Watch** the **Video**, *The Big Picture: I Am, Therefore I Think*, at **MyPsychLab**.

7.1 7.2

Explore the Concept at MyPsychLab

CONCEPT MAP

How People Think

thinking (cognition) refers to mental activities that occur in the brain when processing, organizing, understanding, or communicating information to others

- **mental imagery**
  - mental images are representations for objects or events used in mental activities
  - mental images are interacted with in similar ways as physical objects (e.g., scanning a map or rotating an object)
  - mental images are processed in the brain slightly differently than actual objects
    - as opposed to seeing actual image (eyes → visual cortex → other cortical areas), cortical areas associated with stored knowledge send info to visual cortex
- **concepts**
  - are ideas that represent a class or category of objects, events, or activities
  - are used to interact and organize information without having to think about or process every specific example of the category
  - can represent different levels of objects or events
  - can be well-defined based on strict criteria (formal), or fuzzy, based on personal experience (natural)
  - are represented by prototypes, best examples of the defining characteristics
    - vary according to personal experience, knowledge, and culture
  - are an important tool in problem solving
- **problem solving and decision making**
  - thinking and behaving in certain ways to reach a goal
  - can involve different strategies, logical methods (convergent thinking)
    - **trial and error**
      - trying one solution after another until one works
    - **algorithms**
      - specific, step-by-step procedures for solving certain problems
      - always result in correct solution if there is one
    - **heuristics**
      - simple rules intended to apply to many situations
      - educated guesses based on prior experience
      - generally faster than algorithms but will not always lead to correct solution
  - **insight**
    - "aha!" moments when solution seems to appear in a flash
    - usually based on reorganization of information

PRACTICE quiz How Much Do You Remember?

ANSWERS AVAILABLE IN ANSWER KEY.

Pick the best answer.

1. What is thinking?
- **a.** mental activity that involves processing, organizing, understanding, and communicating information
- **b.** spontaneous, non-directed, and unconscious mental activity
- **c.** simply and succinctly, it is only our ability to remember
- **d.** all mental activity except memory

2. In the brain, the process of creating a mental image is ______________ how we see an actual image.
- **a.** identical to
- **b.** similar to
- **c.** almost the opposite of
- **d.** nothing like

3. People in the United States often think of a sports car when asked to envision a fun, fast form of travel. In this example, a sports car would be considered a
- **a.** prototype.
- **b.** natural concept.
- **c.** formal concept.
- **d.** mental image.

4. What method of problem solving guarantees a solution?
- **a.** heuristic
- **b.** algorithm
- **c.** trial-and-error solution
- **d.** insight

5. What type of problem-solving strategy would be best if you are giving directions to a coffee shop that you currently are at?
- **a.** trial and error
- **b.** availability heuristic
- **c.** representative heuristic
- **d.** work backward from the goal

6. While taking a shower, Miguel suddenly realizes the solution to a problem at work. When later asked how he solved this problem, Miguel said, "The answer just seemed to pop into my head." Miguel's experience is an example of
- **a.** a mechanical solution.
- **b.** a heuristic.
- **c.** an algorithm.
- **d.** insight.

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PROBLEMS WITH PROBLEM SOLVING AND DECISION MAKING

Why does problem solving sometimes fail, and what is meant by creative thinking?

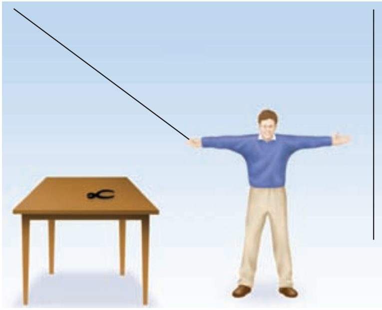

Figure 7.2 The String Problem
How do you tie the two strings together if you cannot reach them both at the same time?

Using insight to solve a problem is not always foolproof. Sometimes a solution to a problem remains just "out of reach" because the elements of the problem are not arranged properly or because people get stuck in certain ways of thinking that act as barriers to solving problems. Such ways of thinking occur more or less automatically, influencing attempts to solve problems without any conscious awareness of that influence. Here's a classic example:

Two strings are hanging from a ceiling but are too far apart to allow a person to hold one and walk to the other. (See **Figure 7.2**.) Nearby is a table with a pair of pliers on it. The goal is to tie the two pieces of string together. How? For the solution to this problem, read on.

People can become aware of automatic tendencies to try to solve problems in ways that are not going to lead to solutions and in becoming aware can abandon the "old" ways for more appropriate problem-solving methods. Three of the most common barriers* to successful problem solving are functional fixedness, mental sets, and confirmation bias.

FUNCTIONAL FIXEDNESS One problem-solving difficulty involves thinking about objects only in terms of their typical uses, which is a phenomenon called **functional fixedness** (literally, "fixed on the function"). Have you ever searched high and low for a screwdriver to fix something around the house? All the while there are several objects close at hand that could be used to tighten a screw: a butter knife, a key, or even a dime in your pocket. Because the tendency is to think of those objects in terms of cooking, unlocking, and spending, we sometimes ignore the less obvious possible uses. The string problem introduced before is an example of functional fixedness. The pair of pliers is often seen as useless until the person realizes it can be used as a weight. (See answer in the section on Creativity.)

Alton Brown, renowned chef and star of the Food Network's *Good Eats* cooking show, is a big fan of what he calls "multitaskers," kitchen items that can be used for more than one purpose. For example, a cigar-cutter can become a tool for cutting carrots, green onions, and garlic. Obviously, Chef Brown is not a frequent victim of functional fixedness.

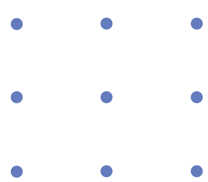

Figure 7.3 The Dot Problem
Can you draw four straight lines so that they pass through all nine dots *without lifting your pencil from the page and without touching any dot more than once?*

MENTAL SETS Functional fixedness is actually a kind of **mental set**, which is defined as the tendency for people to persist in using problem-solving patterns that have worked for them in the past. Solutions that have worked in the past tend to be the ones people try first, and people are often hesitant or even unable to think of other possibilities. Look at **Figure 7.3** and see if you can solve the dot problem.

People are taught from the earliest grades to stay within the lines, right? That tried-and-true method will not help in solving the dot problem. The solution involves drawing the lines beyond the actual dots, as seen in the solution in the section on Creativity.

Answer to insight problem: *Marsha and Marjorie are two of a set of triplets. Gotcha!*

CONFIRMATION BIAS Another barrier to effective decision making or problem solving is **confirmation bias**, the tendency to search for evidence that fits one's beliefs while ignoring any evidence to the contrary. This is similar to a mental set, except that what is "set" is a belief rather than a method of solving problems. Believers in ESP tend to remember the few studies that seem to support their beliefs and psychic predictions that

*barrier: something that blocks one's path; an obstacle preventing a solution.

worked out while at the same time "forgetting" the cases in which studies found no proof or psychics made predictions that failed to come true. They remember only that which confirms their bias toward a belief in the existence of ESP. Another example is that people who believe that they are good multitaskers and can safely drive a motor vehicle while talking or texting on their cell phones may tend to remember their own personal experiences, which may not include any vehicle accidents or "near-misses" (that they are aware of). While it might be tempting to think of one's self as a "supertasker," recent research suggests otherwise. When tested on driving simulators while having to perform successfully on two attention-demanding tasks, over 97 percent of individuals are unable to do so without significant impacts on their performance. During the dual-task condition, only 2.5 percent of individuals were able to perform without problems (Watson & Strayer, 2010). This specific example can be quite dangerous, as it is estimated that at least 28 percent of all traffic crashes are caused by drivers using their cell phone and or texting (National Safety Council, 2010).

The driver of this train was texting from his cell phone immediately before this crash that killed 25 people and injured more than 130 others.

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CREATIVITY

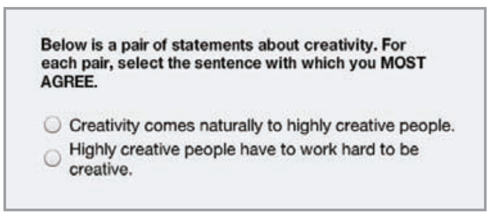
So far, we've only talked about logic and pretty straightforward thinking. How do people come up with totally new ideas, things no one has thought of before?

Not every problem can be answered by using information already at hand and the rules of logic in applying that information. Sometimes a problem requires coming up with entirely new ways of looking at the problem or unusual, inventive solutions. This kind of thinking is called **creativity**: solving problems by combining ideas or behavior in new ways (Csikszentmihalyi, 1996; pronounced chĭck-sĕnt-mē-HĪ-ē). Before we learn more, take the survey experiment *What Is Creativity?* to examine your own beliefs about creativity.

Simulation

What is Creativity?

This survey asks about your experiences with being creative and your beliefs about the creative process and creative people.



Go to the Experiment ▶

Simulate the **Experiment**, *What Is Creativity?*, on **MyPsychLab**

The logical method for problem solving that has been discussed so far is based on a type of thinking called **convergent thinking**. In convergent thinking, a problem is seen as having only one answer and all lines of thinking will eventually lead to (converge on) that single answer by using previous knowledge and logic (Ciardiello, 1998). For example, the question "In what ways are a pencil and a pen alike?" can be answered by listing the

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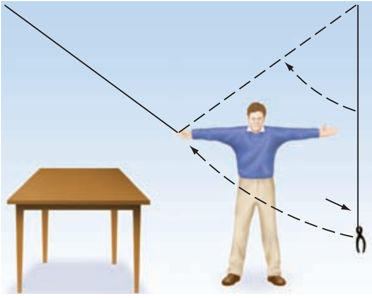


Solution to the String Problem The solution to the string problem is to use the pliers as a pendulum to swing the second string closer to you.

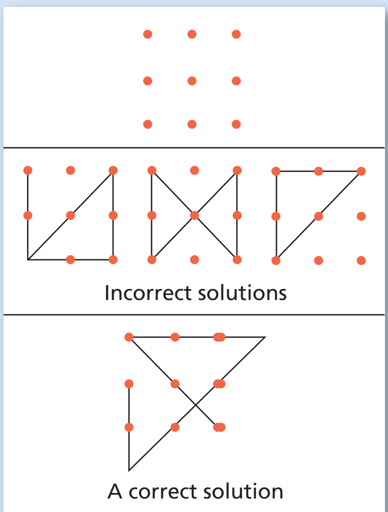
features that the two items have in common: Both can be used to write, have similar shapes, and so on, in a simple comparison process. Convergent thinking works well for routine problem solving but may be of little use when a more creative solution is needed.

Divergent thinking is the reverse of convergent thinking. Here a person starts at one point and comes up with many different, or divergent, ideas or possibilities based on that point (Finke, 1995). For example, if someone were to ask the question, "What is a pencil used for?" the convergent answer would be "to write." But if the question is put this way: "How many different uses can you think of for a pencil?" the answers multiply: "writing, poking holes, a weight for the tail of a kite, a weapon." Divergent thinking has been attributed not only to creativity but also to intelligence (Guilford, 1967).

What are the characteristics of a creative, divergent thinker? Theorists in the field of creative thinking have found through examining the habits of highly creative people that the most productive periods of divergent thinking for those people tend to occur when they are doing some task or activity that is more or less automatic, such as walking or swimming (Csikszentmihalyi, 1996; Gardner, 1993a; Goleman, 1995). These automatic tasks take up some attention processes, leaving the remainder to devote to creative thinking. The fact that all of one's attention is not focused on the problem is actually a benefit, because divergent thinkers often make links and connections at a level of consciousness just below alert awareness, so that ideas can flow freely without being censored* by the higher mental processes (Goleman, 1995). In other words, having part of one's attention devoted to walking, for example, allows the rest of the mind to "sneak up on" more creative solutions and ideas.

Divergent thinkers will obviously be less prone to some of the barriers to problem solving, such as functional fixedness. For example, what would most people do if it suddenly started to rain while they are stuck in their office with no umbrella? How many people would think of using a see-through vinyl tote bag as a makeshift umbrella?

Creative, divergent thinking is often a neglected topic in the education of young people. Although some people are naturally more creative, it is possible to develop one's creative ability. The ability to be creative is important—coming up with topics for a research paper, for example, is something that many students have trouble doing. Cross-cultural research (Basadur et al., 2002; Colligan, 1983) has found that divergent thinking and problem-solving skills cannot be easily taught in the Japanese or Omaha Native American cultures, for example. In these cultures, creativity in many areas is not normally prized and the preference is to hold to well-established, cultural traditions, such as traditional dances that have not varied for centuries. See **Table 7.1** for some ways to become a more divergent thinker.



Solution to the Dot Problem When people try to solve this problem, a mental set causes them to think of the dots as representing a box, and they try to draw the line while staying in the box. The only way to connect all nine dots without lifting the pencil from the paper is to draw the lines so they extend out of the box of dots—literally "thinking outside the box."

Table 7.1

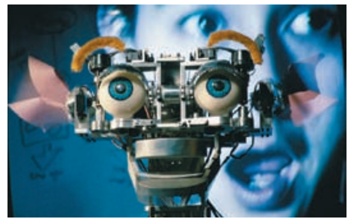
Stimulating Divergent Thinking

Brainstorming	Generate as many ideas as possible in a short period of time, without judging each idea's merits until all ideas are recorded.
Keeping a Journal	Carry a journal to write down ideas as they occur or a recorder to capture those same ideas and thoughts.
Freewriting	Write down or record everything that comes to mind about a topic without revising or proofreading until all of the information is written or recorded in some way. Organize it later.
Mind or Subject Mapping	Start with a central idea and draw a "map" with lines from the center to other related ideas, forming a visual representation of the concepts and their connections.

*censored: blocked from conscious awareness as unacceptable thoughts.

Many people have the idea that creative people are also a little different from other people. There are artists and musicians, for example, who actually encourage others to see them as eccentric. But the fact is that creative people are actually pretty normal. According to Csikszentmihalyi (1997):

1. Creative people usually have a broad range of knowledge about a lot of subjects and are good at using mental imagery.
2. Creative people aren't afraid to be different—they are more open to new experiences than many people, and they tend to have more vivid dreams and daydreams than others do.
3. Creative people value their independence.
4. Creative people are often unconventional in their work, but not otherwise.



Cynthia Breazeal is a researcher at the Artificial Intelligence Lab at M.I.T. Here she is pictured with the robot she designed called Kismet. Designed to help with the study of infant emotional expressions, Kismet can display several "moods" on its face as emotional expressions. This is divergent thinking at its best—a "baby" that won't cry, wet, or demand to be fed.

7.3

Explore the Concept at MyPsychLab

CONCEPT MAP

Problem Solving and Decision Making

- **problems with problem solving**
  - solutions to problems are not always apparent
  - problems can be caused by three common barriers
    - **functional fixedness:** only thinking about objects in terms of their typical uses
    - **mental set:** a tendency to persist in using problem-solving patterns that have worked in the past
    - **confirmation bias:** a tendency to search for evidence that fits your beliefs while ignoring evidence to the contrary
- **creativity**
  - consists of new ways of combining ideas or behavior
  - typically the result of divergent thinking
    - less prone to common barriers of problem solving
    - can be stimulated (see Table 7.1)

Practice quiz How Much Do You Remember? ANSWERS AVAILABLE IN ANSWER KEY.

Pick the best answer.

1. Alicia leaves her office building only to find it is raining. She returns to her office and gets a trash bag out of the supply cabinet. Using a pair of scissors, she cuts the bag so that she can put her head and arms through the bag without getting wet. In using the trash bag as a makeshift rain jacket, Alicia has overcome
a. functional fixedness.
b. confirmation bias.
c. creativity bias.
d. confirmation fixedness.

2. Randall believes that aliens are currently living deep under the ocean. When looking for information about this on the Internet, he ignores any sites that are skeptical of his belief and only visits sites that support his belief. This is an example of
a. functional fixedness.
b. confirmation bias.
c. creativity bias.
d. confirmation fixedness.

3. If you ask someone to give you as many uses as possible for a piece of paper, what type of thinking is being used?
a. divergent
b. functional fixedness
c. convergent
d. mental-set

4. Which of the following is the best way to encourage divergent, creative thinking?
a. Go for a walk or engage in some other automatic activity.
b. Stare at a blank sheet of paper until a new, innovative solution comes to mind.
c. Engage in many activities simultaneously.
d. Force yourself to think of something new and creative.

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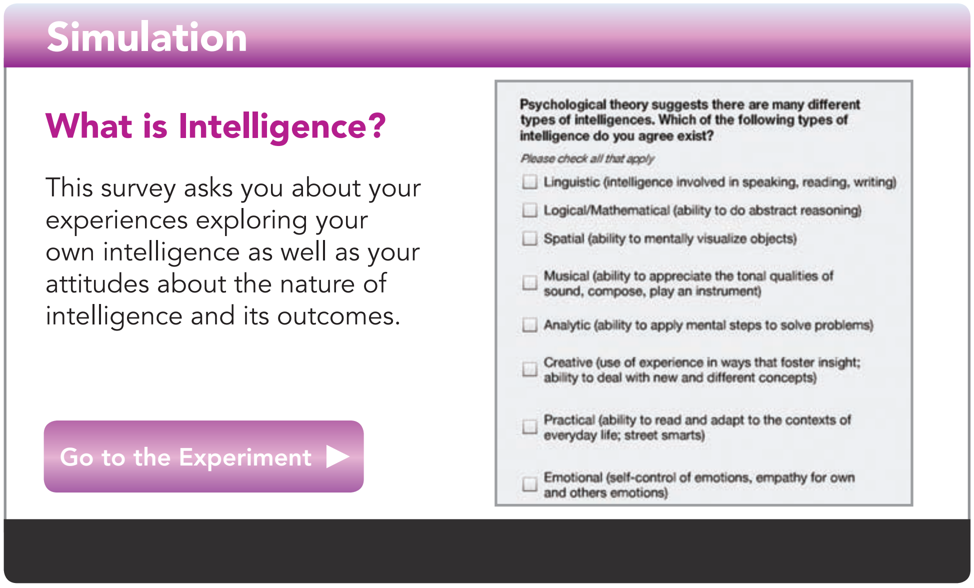
Intelligence

What does it mean to be "smart"? Is this the same as being intelligent? It is likely the answer depends on the immediate task or context. What exactly do we mean by the term *intelligence*? Before we attempt to answer these questions, take the survey experiment *What Is Intelligence?* to discover more about your own notions of intelligence.



Simulate the **Experiment**, *What Is Intelligence?*, on **MyPsychLab**

DEFINITION

How do psychologists define intelligence, and how do various theories of intelligence differ?

Is intelligence merely a score on some test, or is it practical knowledge of how to get along in the world? Is it making good grades or being a financial success or a social success? Ask a dozen people and you will probably get a dozen different answers. Psychologists have come up with a workable definition that combines many of the ideas just mentioned: They define **intelligence** as the ability to learn from one's experiences, acquire knowledge, and use resources effectively in adapting to new situations or solving problems (Sternberg & Kaufman, 1998; Wechsler, 1975). These are the characteristics that individuals need in order to survive in their culture. **Watch** the **Video**, *The Big Picture: What Is Intelligence?* at **MyPsychLab**

THEORIES OF INTELLIGENCE

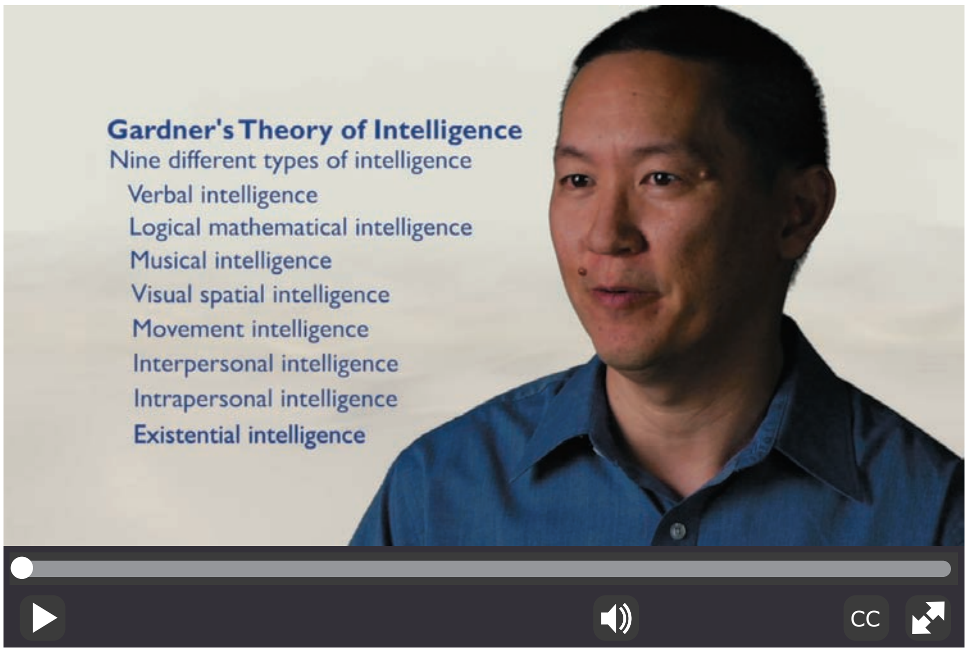
Although we have defined intelligence in a general way, there are differing opinions of the specific knowledge and abilities that make up the concept of intelligence. In the following section, we will discuss three theories that offer different explanations of the nature and number of intelligence-related abilities.

SPEARMAN'S G FACTOR Charles Spearman (1904) saw intelligence as two different abilities. The ability to reason and solve problems was labeled **g factor** for *general intelligence*, whereas task-specific abilities in certain areas such as music, business, or art are labeled **s factor** for *specific intelligence*. A traditional IQ test would most likely measure g factor, but Spearman believed that superiority in one type of intelligence predicts superiority overall. Although his early research found some support for specific intelligences, other researchers (Guilford, 1967; Thurstone, 1938) felt that Spearman had oversimplified the concept of intelligence. Intelligence began to be viewed as composed of numerous factors. In fact, Guilford (1967) proposed that there were 120 types of intelligence.

GARDNER'S MULTIPLE INTELLIGENCES One of the later theorists to propose the existence of several kinds of intelligence is Howard Gardner (1993b, 1999a). Although many people use the terms *reason*, *logic*, and *knowledge* as if they are the same ability, Gardner believes that they are different aspects of intelligence, along with several other abilities. He originally listed seven different kinds of intelligence but later added an eighth type and then proposed a tentative ninth (Gardner, 1998, 1999b). The nine types of intelligence are described in the video *The Basics: Theories of Intelligence*: *Gardner's Theory* and summarized in **Table 7.2**.

The idea of multiple intelligences has great appeal, especially for educators. However, some argue that there are few scientific studies providing evidence for the concept of multiple intelligences (Waterhouse, 2006a, 2006b), while others claim that the evidence does exist (Gardner & Moran, 2006). Some critics propose that such intelligences are no more than different abilities and that those abilities are not necessarily the same thing as what is typically meant by *intelligence* (E. Hunt, 2001).

This child is displaying only one of the many forms that intelligence can take, according to Gardner's multiple intelligences theory.



Watch the **Video**, *The Basics: Theories of Intelligence: Gardner's Theory*, at **MyPsychLab**

Table 7.2
Gardner's Nine Intelligences

TYPE OF INTELLIGENCE	DESCRIPTION	SAMPLE OCCUPATION
Verbal/linguistic	Ability to use language	Writers, speakers
Musical	Ability to compose and/or perform music	Musicians, even those who do not read musical notes but can perform and compose
Logical/mathematical	Ability to think logically and to solve mathematical problems	Scientists, engineers
Visual/spatial	Ability to understand how objects are oriented in space	Pilots, astronauts, artists, navigators
Movement	Ability to control one's body motions	Dancers, athletes
Interpersonal	Sensitivity to others and understanding motivation of others	Psychologists, managers
Intrapersonal	Understanding of one's emotions and how they guide actions	Various people-oriented careers
Naturalist	Ability to recognize the patterns found in nature	Farmers, landscapers, biologists, botanists
Existentialist (a candidate intelligence)	Ability to see the "big picture" of the human world by asking questions about life, death, and the ultimate reality of human existence	Various careers, philosophical thinkers

Source: Gardner, 1998, 1999b.

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Sternberg's practical intelligence is a form of "street smarts" that includes the ability to adapt to one's environment and solve practical problems. These girls are giving their younger brother a drink of water by using a folded leaf as an impromptu cup.

STERNBERG'S TRIARCHIC THEORY Robert Sternberg (1988, 1997b) has theorized that there are three kinds of intelligence. Called the **triarchic theory of intelligence** (*triarchic* means three), this theory includes *analytical*, *creative*, and *practical intelligence*. **Analytical intelligence** refers to the ability to break problems down into component parts, or analysis, for problem solving. This is the type of intelligence that is measured by intelligence tests and academic achievement tests, or "book smarts" as some people like to call it. **Creative intelligence** is the ability to deal with new and different concepts and to come up with new ways of solving problems (divergent thinking, in other words); it also refers to the ability to automatically process certain aspects of information, which frees up cognitive resources to deal with novelty (Sternberg, 2005). **Practical intelligence** is best described as "street smarts," or the ability to use information to get along in life. People with a high degree of practical intelligence know how to be tactful, how to manipulate situations to their advantage, and how to use inside information to increase their odds of success.

How might these three types of intelligence be illustrated? All three might come into play when planning and completing an experiment. For example:

- *Analytical:* Being able to run a statistical analysis on data from the experiment.
- *Creative:* Being able to design the experiment in the first place.
- *Practical:* Being able to get funding for the experiment from donors.

Practical intelligence has become a topic of much interest and research. Sternberg (1996, 1997a, b) has found that practical intelligence predicts success in life but has a surprisingly low relationship to academic (analytical) intelligence. In fact, the higher one's degree of practical intelligence, the less likely that person is to succeed in a university or other academic setting. **Watch** the **Video**, *The Basics: Theories of Intelligence: Sternberg's Theory*, at **MyPsychLab**

MEASURING INTELLIGENCE

How is intelligence measured, how are intelligence tests constructed, and what role do these tests play in neuropsychology?

The history of intelligence testing spans the twentieth century and has at times been marked by controversies and misuse. A full history of how intelligence testing developed would take at least an entire chapter, so this section will discuss only some of the better known forms of testing and how they came to be. **Watch** the **Video**, *Special Topics: Intelligence Testing, Then and Now*, at **MyPsychLab**

It doesn't sound like intelligence would be easy to measure on a test—how do IQ tests work, anyway?

The measurement of intelligence by some kind of test is a concept that is less than a century old. It began when educators in France realized that some students needed more help with learning than others did. They thought that if a way could be found to identify these students more in need, they could be given a different kind of education than the more capable students.

BINET'S MENTAL ABILITY TEST In those early days, a French psychologist named Alfred Binet was asked by the French Ministry of Education to design a formal test of intelligence that would help identify children who were unable to learn as quickly or as well as others, so that they could be given remedial education. Eventually, he and colleague Théodore Simon came up with a test that not only distinguished between fast and slow learners but also between children of different age groups as well (Binet & Simon, 1916). They noticed that the fast learners seemed to give answers to questions that older children might give, whereas the slow learners gave answers that were more typical of a younger child. Binet decided that the key element to be tested was a child's *mental age*, or the average age at which children could successfully answer a particular level of questions.

STANFORD-BINET AND IQ Lewis Terman (1916), a researcher at Stanford University, adopted German psychologist William Stern's method for comparing mental age and *chronological age* (number of years since birth) for use with the translated and revised Binet test. Stern's (1912) formula was to divide the mental age (MA) by the chronological age (CA) and multiply the result by 100 to get rid of any decimal points. The resulting score is called an **intelligence quotient**, or **IQ.** (A *quotient* is a number that results from dividing one number by another.)

$$IQ = MA/CA \times 100$$

For example, if a child who is 10 years old takes the test and scores a mental age of 15 (is able to answer the level of questions typical of a 15-year-old), the IQ would look like this:

$$IQ = 15/10 \times 100 = 150$$

The quotient has the advantage of allowing testers to compare the intelligence levels of people of different age groups. While this method works well for children, it produces IQ scores that start to become meaningless as the person's chronological age passes 16 years. (Once a person becomes an adult, the idea of questions that are geared for a particular age group loses its power. For example, what kind of differences would there be between questions designed for a 30-year-old versus a 40-year-old?) Most intelligence tests today, such as the *Stanford-Binet Intelligence Scales, Fifth Edition* (*SB5*) (Roid, 2003) and the Wechsler tests (see the following section), use age-group comparison norms instead. The SB5 is often used by educators to make decisions about the placement of students into special educational programs, both for those with disabilities and for those with exceptionalities. Many children are given this test in the second grade, or age 7 or 8. The SB5 yields an overall estimate of intelligence, verbal and nonverbal domain scores, all comprised of five primary areas of cognitive ability—fluid reasoning, knowledge, quantitative processing, visual–spatial processing, and working memory (Roid, 2003). See **Table 7.3** for descriptions of some items similar to those from the SB5.

Table 7.3

Paraphrased Sample Items From the Stanford-Binet Intelligence Test

AGE*	TYPE OF ITEM	PARAPHRASED SAMPLE ITEM
2	Board with three differently shaped holes	Child can place correct shape into matching hole on board.
4	Building block bridge	Child can build a simple bridge out of blocks after being shown a model.
7	Similarities	Child can answer such questions as "In what way are a ship and a car alike?"
9	Digit reversal	Child can repeat four digits backward.
Average adult	Vocabulary	Child can define 20 words from a list.

*Age at which item typically is successfully completed.

Source: Roid, G. H. (2003).

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THE WECHSLER TESTS Although the original Stanford-Binet Test is now in its fifth edition and includes different questions for people of different age groups, it is not the only IQ test that is popular today. David Wechsler (Wechsler, 2002, 2003, 2008) was the first to devise a series of tests designed for specific age groups. Originally dissatisfied with the fact that the Stanford-Binet was designed for children but being administered to adults, he developed an IQ test specifically for adults. He later designed tests specifically for older school-age children and preschool children, as well as those in the early grades. The Wechsler Adult Intelligence Scale (WAIS-IV), Wechsler Intelligence Scale for Children (WISC-IV), and the Wechsler Preschool and Primary Scale of Intelligence (WPPSI-IV) are the three versions of this test, and in the United States these tests are now used more frequently than the Stanford-Binet. In earlier editions, another way these tests differed from the Stanford-Binet was by having both a verbal and performance (nonverbal) scale, as well as providing an overall score of intelligence (the original Stanford-Binet was composed predominantly of verbal items). While still using both verbal and nonverbal items, the WISC-IV and WAIS-IV organize items into four index scales that provide an overall score of intelligence and index scores related to four specific cognitive domains—verbal comprehension, perceptual reasoning, working memory, and processing speed. **Table 7.4** has sample items for each of the four index scales from the WAIS-IV.

TEST CONSTRUCTION: GOOD TEST, BAD TEST? All tests are not equally good tests. Some tests may fail to give the same results on different occasions for the same person when that

Table 7.4
Simulated Sample Items From the Wechsler Adult Intelligence Scale (WAIS-IV)

	SIMULATED SAMPLE TEST ITEMS
Verbal Comprehension Index	
Similarities	In what way are a circle and a triangle alike? In what way are a saw and a hammer alike?
Vocabulary	What is a hippopotamus? What does "resemble" mean?
Information	What is steam made of? What is pepper? Who wrote *Tom Sawyer*?
Perceptual Reasoning Index	
Block Design	After looking at a pattern or design, try to arrange small cubes in the same pattern.
Matrix Reasoning	After looking at an incomplete matrix pattern or series, select an option that completes the matrix or series.
Visual Puzzles	Look at a completed puzzle and select three components from a set of options that would re-create the puzzle, all within a specified time limit.
Working Memory Index	
Digit Span	Recall lists of numbers, some lists forward and some lists in reverse order, and recall a mixed list of numbers in correct ascending order.
Arithmetic	Three women divided 18 golf balls equally among themselves. How many golf balls did each person receive? If two buttons cost $0.15, what will be the cost of a dozen buttons?
Processing Speed Index	
Symbol Search	Visually scan a group of symbols to identify specific target symbols, within a specified time limit.
Coding	Learn a different symbol for specific numbers and then fill in the blank under the number with the correct symbol. (This test is timed.)

Simulated items and descriptions similar to those in the *Wechsler Adult Intelligence Scale—Fourth Edition* (2008).

person has not changed—making the test useless. These would be considered unreliable tests. **Reliability** of a test refers to the test producing consistent results each time it is given to the same individual or group of people. For example, if Nicholas takes a personality test today and then again in a month or so, the results should be very similar if the personality test is reliable. Other tests might be easy to use and even reliable, but if they don't actually measure what they are supposed to measure, they are also useless. These tests are thought of as "invalid" (untrue) tests. **Validity** is the degree to which a test actually measures what it's supposed to measure. Another aspect of validity is the extent that an obtained score accurately reflects the intended skill or outcome in real-life situations, or *ecological validity*, not just validity for the testing or assessment situation. For example, we hope that someone who passes his or her test for a driver's license will also be able to safely operate a motor vehicle when they are actually on the road. When evaluating a test, consider what a specific test score means and to what, or to whom, it is compared.

Take the hypothetical example of Professor Stumpwater, who—for reasons best known only to him—believes that intelligence is related to a person's golf scores. Let's say that he develops an adult intelligence test based on golf scores. What do we need to look at to determine if his test is a good one?

Standardization of Tests First of all, we would want to look at how he tried to standardize his test. *Standardization* refers to the process of giving the test to a large group of people that represents the kind of people for whom the test is designed. One aspect of standardization is in the establishment of consistent and standard methods of test administration. All test subjects would take the test under the same conditions. In the professor's case, this would mean that he would have his sample members play the same number of rounds of golf on the same course under the same weather conditions, and so on. Another aspect addresses the comparison group whose scores will be used to compare individual test results. Standardization groups are chosen randomly from the population for whom the test is intended and, like all samples, must be representative of that population. LINK to Learning Objectives A.1 and 1.8. If a test is designed for children, for example, then a large sample of randomly selected children would be given the test.

Norms The scores from the standardization group would be called the *norms*, the standards against which all others who take the test would be compared. Most tests of intelligence follow a *normal curve*, or a distribution in which the scores are the most frequent around the *mean*, or average, and become less and less frequent the further from the mean they occur (see **Figure 7.4**). LINK to Learning Objectives A.2, A.3, and A.4.

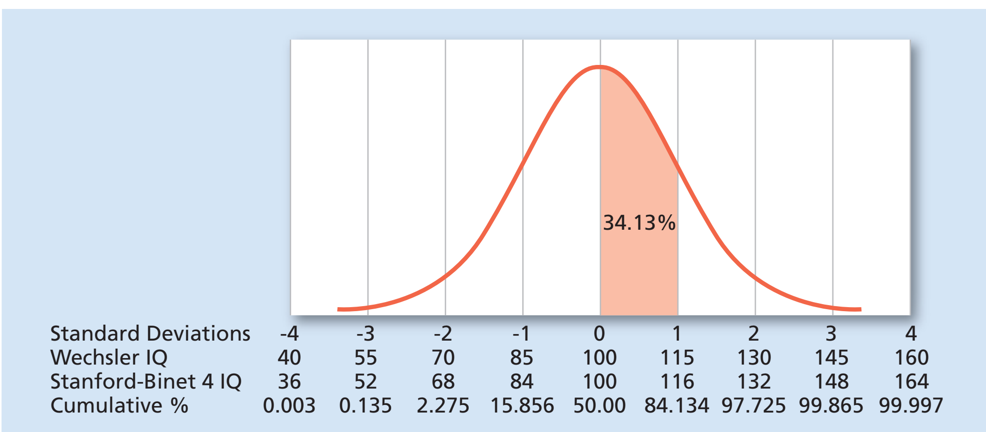

Figure 7.4 The Normal Curve

The percentages under each section of the normal curve represent the percentage of scores falling within that section for each *standard deviation (SD)* from the mean. Scores on intelligence tests are typically represented by the normal curve. The dotted vertical lines each represent one standard deviation from the mean, which is always set at 100. For example, an IQ of 115 on the Wechsler represents one standard deviation above the mean, and the area under the curve indicates that 34.13 percent of the population falls between 100 and 115 on this test. LINK to Learning Objectives A.2, A.3, A.4 and 1.8. Note: The figure shows the mean and standard deviation for the Stanford-Binet Fourth Edition (Stanford-Binet 4). The Stanford-Binet Fifth Edition was published in 2003 and now has a mean of 100 and a standard deviation of 15 for composite scores.

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On the Wechsler IQ test, the percentages under each section of the normal curve represent the percentage of scores falling within that section for each *standard deviation (SD)* from the mean on the test. The standard deviation is the average variation of scores from the mean. LINK to Learning Objective A.4.

In the case of the professor's golf test, he might find that a certain golf score is the average, which he would interpret as average intelligence. People who scored extremely well on the golf test would be compared to the average, as well as people with unusually poor scores.

The normal curve allows IQ scores to be more accurately estimated than the old IQ scoring method formula devised by Stern. Test designers replaced the old ratio IQ of the earlier versions of IQ tests with **deviation IQ scores**, which are based on the normal curve distribution (Eysenck, 1994): IQ is assumed to be normally distributed with a mean IQ of 100 and a typical standard deviation of about 15 (the standard deviation can vary according to the particular test). An IQ of 130, for example, would be two standard deviations above the mean, whereas an IQ of 70 would be two standard deviations below the mean, and in each case the person's score is being compared to the population's average score.

With respect to validity and reliability, the professor's test fares poorly. If the results of the professor's test were compared with other established intelligence tests, there would probably be no relationship at all. Golf scores have nothing to do with intelligence, so the test is not a valid, or true, measure of intelligence.

On the other hand, his test might work well for some people and poorly for others on the question of reliability. Some people who are good and regular golfers tend to score about the same for each game that they play, so for them, the golf score IQ would be fairly reliable. But others, especially those who do not play golf or play infrequently, would have widely varying scores from game to game. For those people, the test would be very unreliable, and if a test is unreliable for some, it's not a good test.

A test can fail in validity but still be reliable. If for some reason Professor Stumpwater chose to use height as a measure of intelligence, an adult's score on Stumpwater's "test" would always be the same, as height does not change by very much after the late teens. But the opposite is not true. If a test is unreliable, how can it accurately measure what it is supposed to measure? For example, adult intelligence remains fairly constant. If a test meant to measure that intelligence gave different scores at different times, it's obviously not a valid measure of intelligence.

Just because an IQ test gives the same score every time a person takes it doesn't mean that the score is actually measuring real intelligence, right?

That's right—think about the definition of intelligence for a moment: the ability to learn from one's experiences, acquire knowledge, and use resources effectively in adapting to new situations or solving problems. How can anyone define what "effective use of resources" might be? Does everyone have access to the same resources? Is everyone's "world" necessarily perceived as being the same? Intelligence tests are useful measuring devices but should not necessarily be assumed to be measures of all types of intelligent behavior, or even good measures for all groups of people, as the next section discusses.

How might these two women, apparently from different cultures, come to an agreement on what best defines intelligence?

IQ TESTS AND CULTURAL BIAS The problem with trying to measure intelligence with a test that is based on an understanding of the world and its resources is that not everyone comes from the same "world." People raised in a different culture, or even a different economic situation, from the one in which the designer of an IQ test is raised are not likely to perform well on such a test—not to mention the difficulties of taking a test that is written in an unfamiliar language or dialect. In the early days of immigration, people from non-English-speaking countries would score very poorly on intelligence tests, in some cases being denied entry to the United States on the basis of such tests (Allen, 2006).

It is very difficult to design an intelligence test that is completely free of *cultural bias*, a term referring to the tendency of IQ tests to reflect, in language, dialect, and content, the culture of the person or persons who designed the test. A person who comes from the same culture (or even socioeconomic background) as the test designer may have an unfair advantage over a person who is from a different cultural or socioeconomic background (Helms, 1992). If people raised in an Asian culture are given a test designed within a traditional Western culture, many items on the test might make no sense to them. For example, one kind of question might be: Which one of the five is least like the other four?

DOG—CAR—CAT—BIRD—FISH

The answer is supposed to be "car," which is the only one of the five that is not alive. But a Japanese child, living in a culture that relies on the sea for so much of its food and culture, might choose "fish," because none of the others are found in the ocean. That child's test score would be lower but not because the child is not intelligent.

In 1971, Adrian Dove designed an intelligence test to highlight the problem of cultural bias. Dove, an African American sociologist, created the Dove Counterbalance General Intelligence Test (later known as the Chitling Test) in an attempt to demonstrate that a significant language/dialect barrier exists among children of different backgrounds. Questions on this test were derived from the African-American culture and asked for information that would not be knowledge readily available to non-African-Americans. For example, one question asks about the proper length of time for cooking chitlings (the answer is 24 hours, just so you know).

Anyone not part of the African American culture of the southeastern United States in the 1960s and 1970s will probably score very poorly on this test, including African American people from different geographical regions. The point is simply this: Tests such as these are created by people who are from a particular culture and background. Test questions and answers that the creators might think are common knowledge may relate to their own experiences and not to people of other cultures, backgrounds, or socioeconomic levels.

Attempts have been made to create intelligence tests that are as free of cultural influences as is humanly possible. Many test designers have come to the conclusion that it may be impossible to create a test that is completely free of cultural bias (Carpenter et al., 1990). Instead, they are striving to create tests that are at least *culturally fair*. These tests use questions that do not create a disadvantage for people whose culture differs from that of the majority. Many items on a "culture-fair" test require the use of nonverbal abilities, such as rotating objects, rather than items about verbal knowledge that might be culturally specific.

If intelligence tests are so flawed, why do people still use them?

USEFULNESS OF IQ TESTS IQ tests are generally valid for predicting academic success and job performance (Sackett et al., 2008). This may be more true for those who score at the higher and lower ends of the normal curve. (For those who score in the average range of IQ, the predictive value is less clear.) The kinds of tests students are given in school are often similar to intelligence tests, and so people who do well on IQ tests typically do well on other kinds of academically oriented tests as well, such as the SAT, the

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American College Test (ACT), the Graduate Record Exam (GRE), and actual college examinations. These achievement tests are very similar to IQ tests but are administered to groups of people rather than to individuals. However, recent research suggests skills in self-regulation or levels of motivation may impact IQ measures and raises concerns about situations or circumstances where IQ scores may not be unbiased predictors of academic or job success (Duckworth et al., 2011; Duckworth & Seligman, 2005; Nisbett et al., 2012). **Watch** the **Video**, *Thinking Like a Psychologist: Intelligence Tests and Success*, at **MyPsychLab**

Intelligence testing also plays an important role in neuropsychology, where specially trained psychologists use intelligence tests and other forms of cognitive and behavioral testing to assess neurobehavioral disorders in which cognition and behavior are impaired as the result of brain injury or brain malfunction (National Academy of Neuropsychology, 2001). As part of their profession, neuropsychologists use intelligence testing in diagnosis (e.g., head injury, learning disabilities, neuropsychological disorders), tracking progress of individuals with such disorders, and in monitoring possible recovery. For more on neuropsychological assessment, see the Psychology in the News section.

psychology in the news

Neuropsychology Sheds Light on Head Injuries

Many of the topics in this chapter are related to the interests of cognitive psychologists, cognitive neuroscientists, and neuropsychologists alike, but here we will expand on the work of clinical neuropsychologists LINK to Learning Objective B.5, who often work with individuals who have traumatic brain injury (TBI). Unlike a broken limb or other bodily injury that might result in a temporary loss of function, many traumatic brain injuries not only have immediate effects but can also be permanent, impacting the day-to-day functioning of both individuals and their loved ones for the rest of their lives. Depending on the area or areas of the brain injured and the severity of the trauma, some possible outcomes might include difficulty thinking, speech disturbances, memory problems, reduced attention span, headaches, sleep disturbances, frustration, mood swings, and personality changes. Not only do these outcomes negatively impact formal tests of intelligence, the deficits from such injuries may also affect thinking, problem solving, and cognition in general.

Mild traumatic brain injury, or concussion, is an impairment of brain function for minutes to hours following a head injury. Concussions may include a loss of consciousness for up to 30 minutes, "seeing stars," headache, dizziness, and sometimes nausea or vomiting (Blumenfeld, 2011; Ruff et al., 2009). Amnesia for the events immediately before or after the accident is also a primary symptom and more likely to be anterograde in nature. LINK to Learning Objective 6.5 and 6.12. With regard to concussions and other levels of traumatic brain injury, athletes and military personnel have been of particular interest to neuropsychologists, as they have been a vehicle for new findings about different types of injuries and the effects of repeated injury on long-term outcomes.

Athletics In high school athletes, concussions account for approximately 9 percent of all high school sports-related injuries; a recent survey of 15 college-level sports over a 16-year period found the rate of concussions has increased significantly (Gessel et al., 2007; Hootman et al., 2007). Cheerleading also has its share of head injuries, as concussions are among the five most common injuries reported in a sample of 412 cheerleading teams ranging from elementary school to college (Shields & Smith, 2009).

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The effects of repeated concussions and the long-term effects of head injuries in general are of particular interest to neuropsychologists and other health professionals because the potential issues (memory problems, changes in personality, etc.) may not be evident until many years later. American football is one sport in which athletes may have extended playing careers. The possibility of an increased risk for depression, dementia, or other neurological risks for these athletes after they have quit playing has spawned ongoing research with professional football players (Guskiewicz et al., 2007; G. Miller, 2009; Hazrati et al., 2013). Former players who had three or more concussions were 3 times more likely to have significant memory problems and 5 times more likely to be diagnosed with mild cognitive impairment, often a precursor to Alzheimer's disease. Additional research suggests that some players develop high concentrations of the protein tau (LINK to Learning Objective 6.12), which has also been associated with Alzheimer's disease (Guskiewicz et al., 2005; McKee et al., 2009). Increased body weight is also being investigated as a dangerous combination for professional football players, as this can interfere with normal blood brain flow and may increase the negative consequences of repeated brain trauma (Willeumier et al., 2012). Researchers are also working to determine if there is something unique in the type of brain pathology that results in dementia as the result of a traumatic brain injury (TBI) when compared to the type of neuronal degeneration that results in Alzheimer's disease--in addition to professional athletes, individuals in the military are also at higher risk of TBI-related dementia (Goldstein et al., 2012; Shively et al., 2012).

Military Historically, many military conflicts have been associated with a "signature wound," which is an injury that is suffered by a substantial number of veterans from that particular war. The wound may be physical or psychological in nature. For instance, "shell shock" is often associated with many veterans of World War I. For the Vietnam War, post-traumatic stress disorder is the pervasive injury that comes to mind. In the ongoing conflicts in Iraq and Afghanistan, the signature wound may be TBI (E. Jones et al., 2007; Okie, 2005). The degree of brain injuries being sustained range from mild to moderate to severe, and over 50 percent are considered to be moderate to severe (Okie, 2005). In some studies, more than 15 percent of soldiers returning from Iraq report experiencing a mild traumatic brain injury, most likely the result of high intensity combat or a blast mechanism (Hoge et al., 2008). Many of these blast injuries are caused by IEDs, or "improvised explosive devices." The prevalence of IEDs is currently greater in Iraq than it is in Afghanistan, with troops in Iraq being approximately 1.7 times more likely to be hospitalized with traumatic brain injury. Unfortunately, it is a trend that appears to be increasing (Wojcik et al., 2010). The pervasiveness of IEDs in Iraq has generated new areas of research with the goal of improving the lives of the injured by understanding the unique outcomes and consequences associated with this particular type of head injury, as it appears to impact the brain in ways not seen in other types of head injury. For example, both resting state and specific task-based fMRI protocols may prove beneficial in understanding the brain and functional changes that occur in blast TBI (Graner et al., 2013).

To learn more about traumatic brain injury:

National Institute of Neurological Disorders and Stroke **www.ninds.nih.gov/disorders/tbi/tbi.htm**

To learn more about blast injuries:

Centers for Disease Control and Prevention
www.bt.cdc.gov/masscasualties/explosions.asp

Questions for Further Thought

1. Do you know someone with a TBI? How has the injury affected his or her life?
2. Who do you think has a better chance of recovery from a TBI, a child or and adult? Why?

7.4 7.5

Explore the Concept at MyPsychLab

CONCEPT MAP

Intelligence

(the ability to learn from one's experiences, acquire knowledge, and use resources effectively)

- **theories**
  - **Spearman's g factor:** intelligence comprises two different abilities
    - **g factor:** general intelligence
    - **s factor:** specific intelligence
  - **Gardner's multiple intelligences:** overall intelligence comprises nine different types
    - (see Table 7.2)
  - **Sternberg's triarchic theory:** intelligence comprises three different aspects
    - **analytical**
    - **creative**
    - **practical**

Measuring Intelligence

- first formal test created by Alfred Binet and Theodore Simon to help identify French students who needed more help with learning
- **tests**
  - **Binet's Mental Ability Test**
    - key element to be tested was child's mental age
  - **Stanford-Binet**
    - **Terman** (researcher at Stanford) translated and revised Binet's test
    - first test to adopt intelligence quotient (IQ): IQ = mental age/chronological age × 100
      - now uses age-group comparison norms as the Wechsler does
    - uses a variety of verbal and nonverbal subtests to provide an overall estimate of intelligence and scores related to five areas of cognition (see Table 7.3)
  - **Wechsler Tests**
    - uses a variety of verbal and performance subtests to provide an overall score of intelligence and index scores related to four specific cognitive domains (see Table 7.4)
- **test construction**
  - good tests are both valid and reliable
  - standardized administration, scoring, and comparison against norms
  - intelligence is assumed to follow a normal curve
  - is challenging
    - different definitions of intelligence and multiple ways to assess them
    - difficult to design tests that are completely free of cultural bias

PRACTICE quiz How Much Do You Remember?

ANSWERS AVAILABLE IN ANSWER KEY.

Pick the best answer.

1. In Gardner's view, effective counseling psychologists and managers would likely be high in __________ intelligence.
- **a.** verbal/linguistic
- **b.** visual–spatial
- **c.** interpersonal
- **d.** intrapersonal

2. According to Sternberg, which type of intelligence has a low relationship to academic success and would be the most difficult to measure in the classroom?
- **a.** practical
- **b.** creative
- **c.** analytical
- **d.** verbal

3. By what age do IQ scores start to become meaningless?
- **a.** 5
- **b.** 10
- **c.** 16
- **d.** 30

4. Liv is 4 years old. The intelligence test that would most likely be used to determine her IQ is the
- **a.** WAIS-IV.
- **b.** WISC-IV.
- **c.** WPPSI-IV.
- **d.** Dove Test.

5. Professor Becker designed an IQ test. To validate this test, the professor should be careful to do which of the following?
- **a.** Give the test at least twice to the same group to ensure accuracy.
- **b.** Select the people in the sample from the population of people for whom the test is designed.
- **c.** Select only university professors to take the test so that they can critique the questions on the test.
- **d.** Strive to make sure that the test measures what it is supposed to measure.

6. In terms of differing cultures, what should be the goal of every test designer?
- **a.** to create a test free of cultural bias
- **b.** to create a test that is culturally fair
- **c.** to create a test with no questions involving culture
- **d.** to create a series of culture-varied tests

THINKING CRITICALLY:
What kind of questions would you include on an intelligence test to minimize cultural bias?

EXTREMES OF INTELLIGENCE

Another use of IQ tests is to help identify people who differ from those of average intelligence by a great degree. Although one such group is composed of those who are sometimes called "geniuses" (who fall at the extreme high end of the normal curve for intelligence), the other group is made up of people who, for various reasons, are considered intellectually disabled and whose IQ scores fall well below the mean on the normal curve.

INTELLECTUAL DISABILITY

What is intellectual disability and what are its causes?

Intellectual disability (intellectual developmental disorder) (formerly *mental retardation* or *developmentally delayed*) is a neurodevelopmental disorder and is defined in several ways. First, the person exhibits deficits in mental abilities, which is typically associated with an IQ score approximately two standard deviations below the mean on the normal curve, such as below 70 on a test with a mean of 100 and standard deviation of 15. Second, the person's *adaptive behavior* (skills that allow people to live independently, such as being able to work at a job, communicate well with others, and grooming skills such as being able to get dressed, eat, and bathe with little or no help) is severely below a level appropriate for the person's age. Finally, these limitations must begin in the developmental period. Intellectual disability occurs in about 1 percent of the population (American Psychiatric Association, 2013).

So how would a professional go about deciding whether or not a child has an intellectual disability? Is the IQ test the primary method?

Diagnosis Previous editions of the *Diagnostic and Statistical Manual of Mental Disorders (DSM)* relied heavily on IQ tests for determining the diagnosis of *mental retardation* and level of severity. This has changed with the release of the newest edition in 2013, the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)* (American Psychiatric Association, 2013) and is consistent with recommendations from the American Association on Intellectual and Developmental Disabilities (AAIDD) (AAIDD, 2009; Schalock et al., 2010). Recognizing tests of IQ are less valid as one approaches the lower end of the IQ range, and the importance of adaptive living skills in multiple life areas, levels of severity are now based on level of adaptive functioning and level of support the individual requires (American Psychiatric Association, 2013). Thus, a *DSM-5* diagnosis of intellectual disability is based on deficits in intellectual functioning, determined by standardized tests of intelligence and clinical assessment, which impact adaptive functioning across three domains. The domains include: conceptual (memory, reasoning, language, reading, writing, math, and other academic skills), social (empathy, social judgement, interpersonal communication, and other skills that impact the ability to make and maintain friendships), and practical (self-management skills that affect personal care, job responsibilities, school, money management, and other areas) (American Psychiatric Association, 2013). Previous editions indicated these deficits must occur prior to 18 years of age, but the *DSM-5* removes the specific age criteria, specifying symptoms must begin during the developmental period.

Intellectual disability can vary from mild to profound. According to the *DSM-5* (American Psychiatric Association, 2013), individuals with mild intellectual disability may not be recognized as having deficits in the conceptual domain until they reach school age where learning difficulties become apparent; as an adult, they are likely to be fairly concrete thinkers. In the social domain, they are at risk of being manipulated as social judgment and interactions are immature as compared to same-age peers. In the practical

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This middle-aged man, named Jack, lives in a small town in Arkansas and serves as a deacon in the local church. He is loved and respected and leads what, for him, is a full and happy life. Jack also has Down syndrome but he has managed to find his place in the world.

domain, they are capable of living independently with proper supports in place but will likely require assistance with more complex life skill such as health care decisions, legal issues, or raising a family (American Psychiatric Association, 2013). This category makes up the vast majority of those with intellectual disabilities. Other classifications in order of severity are moderate, severe, and profound. Conceptually, individuals with profound intellectual disability have a very limited ability to learn beyond simple matching and sorting tasks and socially, have very poor communication skills, although they may recognize and interact nonverbally with well-known family members and other caretakers. In the practical domain, they may be able to participate by watching or assisting, but are likely totally dependent upon on others for all areas of their care (American Psychiatric Association, 2013). All of these skill deficits are likely compounded by multiple physical or sensory impairments.

Causes What causes intellectual disability? Unhealthy living conditions can affect brain development. Examples of such conditions are lead poisoning from eating paint chips (Lanphear et al., 2000), exposure to PCBs (Darvill et al., 2000), prenatal exposure to mercury (Grandjean et al., 1997), as well as other toxicants (Ericksson et al., 2001; Eskenazi et al., 1999; Schroeder, 2000). Deficits may also be attributed to factors resulting in inadequate brain development or other health risks associated with poverty. Examples include malnutrition, health consequences as the result of not having adequate access to health care, or lack of mental stimulation through typical cultural and educational experiences.

Some of the biological causes of intellectual disability include Down syndrome (LINK to Learning Objective 8.3), fetal alcohol syndrome, and fragile X syndrome. *Fetal alcohol syndrome* is a condition that results from exposing a developing embryo to alcohol, and intelligence levels can range from below average to levels associated with intellectual disability (Olson & Burgess, 1997). In *fragile X syndrome*, a male has a defect in a gene on the X chromosome of the 23rd pair, leading to a deficiency in a protein needed for brain development. Depending on the severity of the damage to this gene, symptoms of fragile X syndrome can range from mild to severe or profound intellectual disability (Dykens et al., 1994; Valverde et al., 2007).

There are many other causes of intellectual disability (Murphy et al., 1998). Lack of oxygen at birth, damage to the fetus in the womb from diseases, infections, or drug use by the mother, and even diseases and accidents during childhood can lead to intellectual disability.

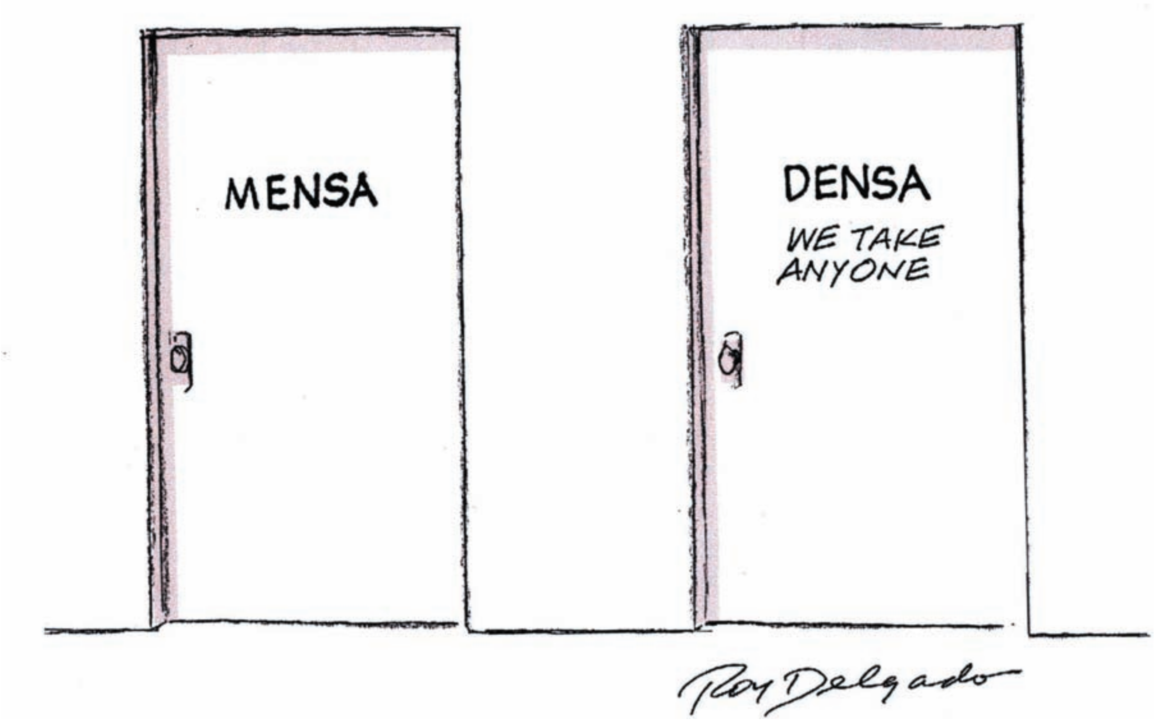
One thing should always be remembered: Intellectual disability affects a person's *intellectual* capabilities and adaptive behaviors. Individuals with an intellectual disability are just as responsive to love and affection as anyone else and need to be loved and to have friends just as all people do. Intelligence is only one characteristic; warmth, friendliness, caring, and compassion also count for a great deal and should not be underrated.

GIFTEDNESS

What defines giftedness, and how are giftedness and emotional intelligence related to success in life?

At the other end of the intelligence scale* are those who fall on the upper end of the normal curve (see Figure 7.4), above an IQ of 130 (about 2 percent of the population). The term applied to these individuals is **gifted**, and if their IQ falls above 140 to 145 (less than half of 1 percent of the population), they are often referred to as highly advanced or *geniuses*.

*scale: a graded series of tests or performances used in rating individual intelligence or achievement.



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I've heard that geniuses are sometimes a little "nutty" and odd. Are geniuses, especially the really high-IQ ones, "not playing with a full deck," as the saying goes?

People have long held many false beliefs about people who are very, very intelligent. Such beliefs have included that gifted people are weird and socially awkward, physically weak, and more likely to suffer from mental illnesses. From these beliefs come the "mad scientist" of the cinema and the "evil geniuses" of literature.

These beliefs were shattered by a groundbreaking study that was initiated in 1921 by Lewis M. Terman, the same individual responsible for the development of the Stanford-Binet Test. Terman (1925) selected 1,528 children to participate in a longitudinal study. LINK to Learning Objective 8.1. These children, 857 boys and 671 girls, had IQs (as measured by the Stanford-Binet) ranging from 130 to 200. The early findings of this major study (Terman & Oden, 1947) demonstrated that the gifted were socially well adjusted and often skilled leaders. They were also above average in height, weight, and physical attractiveness, putting an end to the myth of the weakling genius. Terman was able to demonstrate not only that his gifted children were *not* more susceptible to mental illness than the general population, but he was also able to show that they were actually more resistant to mental illnesses than those of average intelligence. Only those with the highest IQs (180 and above) were found to have some social and behavioral adjustment problems *as children* (Janos, 1987).

Stanford University psychologist Lewis Terman is pictured at his desk in 1942. Terman spent a good portion of his career researching children with high IQ scores and was the first to use the term *gifted* to describe these children.

Terman's "Termites," as they came to be called, were also typically successful as adults. They earned more academic degrees and had higher occupational and financial success than their average peers (at least, the men in the study had occupational success—women at this time did not typically have careers outside the home). Researchers Zuo and Cramond (2001) examined some of Terman's gifted people to see if their identity formation as adolescents was related to later occupational success. LINK to Learning Objective 8.8. They found that most of the more successful "Termites" had in fact successfully achieved a consistent sense of self, whereas those who were less successful had not done so. For more on Terman's famous study, see Classic Studies in Psychology.

classic studies in psychology

Terman's "Termites"

Terman's (1925) longitudinal study is still going on today, although many of his original subjects have passed away and those who remain are in their 90s. Terman himself died in 1956, but several other researchers (including Robert Sears, one of the original "Termites") have kept track of the remaining "Termites" over the years (Holahan & Sears, 1996).

As adults, the "Termites" were relatively successful, with a median income in the 1950s of $10,556, compared to the national median at that time of $5,800 a year. Most of them graduated from college, many earning advanced degrees. Their occupations included doctors, lawyers, business executives, university professors, scientists, and even one famous science fiction writer and an Oscar-winning director (Edward Dmytryk, director of *The Caine Mutiny* in 1954, among others).

By 2000, only about 200 "Termites" were still living. Although the study was marred by several flaws, it still remains one of the most important and rich sources of data on an entire generation. Terman's study was actually the first truly longitudinal study (LINK to Learning Objective 8.1) ever to be accomplished, and scientists have gotten data about the effects of phenomena such as World War II and the influence of personality traits on how long one lives from the questionnaires filled out by the participants over the years.

Terman and Oden (1959) compared the 100 most successful men in the group to the 100 least successful by defining "successful" as holding jobs that related to or used their intellectual skills. The more successful men earned more money, had careers with more prestige, and were healthier and less likely to be divorced or alcoholics than the less successful men. The IQ scores were relatively equal between the two groups, so the differences in success in life had to be caused by some other factor or factors. Terman and Oden found that the successful adults were different from the others in three ways: They were more goal oriented, more persistent in pursuing those goals, and were more self-confident than the less successful "Termites."

What were the flaws in this study? Terman acquired his participants by getting recommendations from teachers and principals, not through random selection, so that there was room for bias in the pool of participants from the start. It is quite possible that the teachers and principals were less likely, especially in 1921, to recommend students who were "troublemakers" or different from the majority. Consequently, Terman's original group consisted of almost entirely White, urban, and middle-class children, with the majority (857 out of 1,528) being male. There were only 2 African Americans, 6 Japanese Americans, and 1 Native American.

Another flaw is the way Terman interfered in the lives of his "children." In any good research study, the investigator should avoid becoming personally involved in the lives of the participants in the study to reduce the possibility of biasing the results. Terman seemed to find it nearly impossible to remain objective (Leslie, 2000). He became like a surrogate father to many of them.

Flawed as it may have been, Terman's groundbreaking study did accomplish his original goal of putting to rest the myths that existed about genius in the early part of the twentieth century. Gifted children and adults are no more prone to mental illnesses or odd behavior than any other group, and they also have their share of failures as well as successes. Genius is obviously not the only factor that influences success in life—personality and experiences are strong factors as well. For example, the homes of the children in the top 2 percent of Terman's group had an average of 450 books in their libraries, a sign that the parents of these children valued books and learning, and these parents were also more likely to be teachers, professionals, doctors, and lawyers. The experiences of these gifted children growing up would have been vastly different from those in homes with less emphasis on reading and lower occupational levels for the parents.

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Questions for Further Discussion

1. In Terman and Oden's 1959 study of the successful and unsuccessful "Termites," what might be the problems associated with the definition of "successful" in the study?
2. Thinking back to the discussion of research ethics in Chapter One (LINK to Learning Objective 1.13), what ethical violations may Terman have committed while involved in this study?
3. If gifted children thrive when growing up in more economically sound and educationally focused environments, what should the educational system strive to do to nourish the gifted? Should the government get involved in programs for the gifted?

A book by Joan Freeman called *Gifted Children Grown Up* (Freeman, 2001) describes the results of a similar longitudinal study of 210 gifted and nongifted children in Great Britain. One of the more interesting findings from this study is that gifted children who are "pushed" to achieve at younger and younger ages, sitting for exams long before their peers would do so, often grow up to be disappointed, somewhat unhappy adults. Freeman points to differing life conditions for the gifted as a major factor in their success, adjustment, and well-being: Some lived in poverty and some in wealth, for example. Yet another longitudinal study (Torrance, 1993) found that in both gifted students and gifted adults there is more to success in life than intelligence and high academic achievement. In that study, liking one's work, having a sense of purpose in life, a high energy level, and persistence were also very important factors. If the picture of the genius as mentally unstable is a myth, so, too, is the belief that being gifted will always lead to success, as even Terman found in his original study.

EMOTIONAL INTELLIGENCE What about people who have a lot of "book smarts" but not much common sense? There are some people like that, who never seem to get ahead in life, in spite of having all that so-called intelligence. It is true that not everyone who is intellectually able is going to be a success in life (Mehrabian, 2000). Sometimes the people who are most successful are those who didn't do all that well in the regular academic setting.

One explanation for why some people who do poorly in school succeed in life and why some who do well in school don't do so well in the "real" world is that success relies on a certain degree of **emotional intelligence**, the accurate awareness of and ability to manage one's own emotions to facilitate thinking and attain specific goals, and the ability to understand what others feel (Mayer & Salovey, 1997; Mayer, Salovey, et al., 2008).

The concept of emotional intelligence was first introduced by Peter Salovey and John Mayer (1990) and later popularized by Dan Goleman (1995). And while Goleman originally suggested emotional intelligence was a more powerful influence on success in life than more traditional views of intelligence, his work and the work of others used the term in a variety of different ways than originally proposed, and claims by some were not backed by scientific evidence. For example, emotional intelligence is not the same as having high self-esteem or being optimistic. One who is emotionally intelligent possesses self-control of emotions such as anger, impulsiveness, and anxiety. Empathy, the ability to understand what others feel, is also a component, as are an awareness of one's own emotions, sensitivity, persistence even in the face of frustrations, and the ability to motivate oneself (Salovey & Mayer, 1990; Mayer & Salovey, 1997).

That all sounds very nice, but how can anything like this be measured?

Is there research to support this idea? In one study, researchers asked 321 participants to read passages written by nonparticipants and try to guess what the

Emotional intelligence includes empathy, which is the ability to feel what others are feeling. This doctor is not only able to listen to her patient's problems but also is able to show by her facial expression, body language, and gestures that she understands how the patient feels.

nonparticipants were feeling while they were writing (Mayer & Geher, 1996). The assumption was that people who were good at connecting thoughts to feelings would also have a high degree of empathy and emotional intelligence. The participants who more correctly judged the writers' emotional experiences (assessed by both how well each participant's emotional judgments agreed with a group consensus and the nonparticipant's actual report of feelings) also scored higher on the empathy measure and lower on the defensiveness measure. These same participants also had higher SAT scores (self-reported), leading Mayer and colleagues to conclude not only that emotional intelligence is a valid and measurable concept but also that general intelligence and emotional intelligence may be related: Those who are high in emotional intelligence are also smarter in the traditional sense (Mayer et al., 2000). A more recent review found individuals with higher emotional intelligence tended to have better social relationships for both children and adults, better family and intimate relationships, were perceived more positively by others, had better academic achievement, were more successful at work, and experienced greater psychological well-being (Mayer, Roberts, et al., 2008).

THE NATURE/NURTURE CONTROVERSY REGARDING INTELLIGENCE

What is the influence of heredity and environment on the development of intelligence?

Are people born with all of the "smarts" they will ever have, or does experience and learning count for something in the development of intellect? The influence of nature (heredity or genes) and nurture (environment) on personality traits has long been debated in the field of human development, and intelligence is one of the traits that has been examined closely. LINK to Learning Objective 8.2. **Watch** the **Video**, *What's in It for Me?: How Resilient Are You?*, at **MyPsychLab**

TWIN AND ADOPTION STUDIES The problem with trying to separate the role of genes from that of environment is that controlled, perfect experiments are neither practical nor ethical. Instead, researchers find out what they can from *natural experiments*, circumstances existing in nature that can be examined to understand some phenomenon. *Twin studies* are an example of such circumstances.

Identical twins are those who originally came from one fertilized egg and, therefore, share the same genetic inheritance. Any differences between them on a certain trait, then, should be caused by environmental factors. Fraternal twins come from two different eggs, each fertilized by a different sperm, and share only the amount of genetic material that any two siblings would share. LINK to Learning Objective 8.3. By comparing the IQs of these two types of twins reared together (similar environments) and reared apart (different environments), as well as persons of other degrees of relatedness, researchers can get a general, if not exact, idea of how much influence heredity has over the trait of intelligence (see **Figure 7.5**). As can be easily seen from the chart, the greater the degree of genetic relatedness, the stronger the correlation is between the IQ scores of those persons. The fact that genetically identical twins show a correlation of 0.86 means that the environment must play a part in determining some aspects of intelligence as measured by IQ tests. If heredity alone were responsible, the correlation between genetically identical twins should be 1.00. At this time, researchers have determined that the estimated **heritability** (proportion of change in IQ within a population that is caused by hereditary factors) for intelligence is about .50 or 50 percent (Plomin & DeFries, 1998; Plomin & Spinath, 2004). Furthermore, the impact of genetic factors increases with increasing age, but the set of genes or genetic factors remain the same. The effects of the same set of genes becomes larger with increasing age (Posthuma et al., 2009).

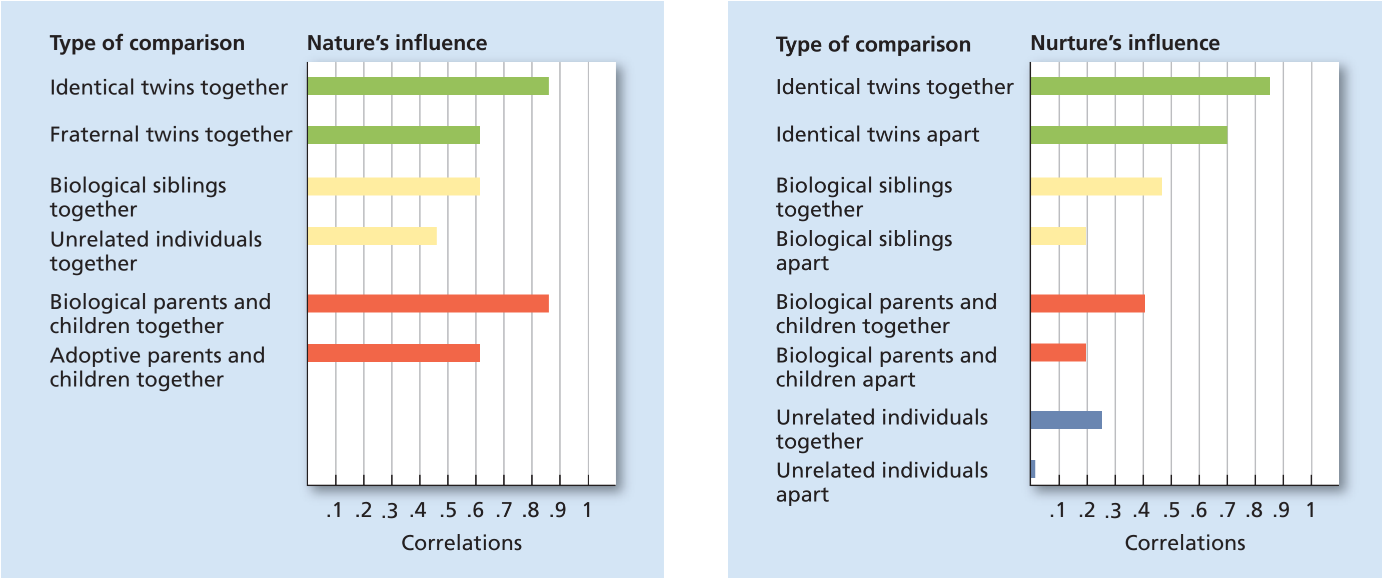

Figure 7.5 Correlations Between IQ Scores of Persons With Various Relationships

In the graph on the left, the degree of genetic relatedness seems to determine the agreement (correlation) between IQ scores of the various comparisons. For example, identical twins, who share 100 percent of their genes, are more similar in IQ than fraternal twins, who share only about 50 percent of their genes, even when raised in the same environment. In the graph on the right, identical twins are still more similar to each other in IQ than are other types of comparisons, but being raised in the same environment increases the similarity considerably.

7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9 7.10 7.11

Wait a minute—if identical twins have a correlation of .86, wouldn't that mean that intelligence is 86 percent inherited?

Although the correlation between identical twins is higher than the estimated heritability of .50, that similarity is not entirely due to the twin's genetic similarity. Twins who are raised in the same household obviously share very similar environments as well. Even twins who are reared apart, as seen in adoption studies, are usually placed in homes that are similar in socioeconomic and ethnic background—more similar than one might think. So when twins who are genetically similar are raised in similar environments, their IQ scores are also going to be similar. However, similar environmental influences become less important over time (where genetic influences increase over time), accounting for only about 20 percent of the variance in intelligence by age 11 or 12 (Posthuma et al., 2009). In turn, environmental influences tend not to be a factor by adolescence, and with the increasing impact of genetic factors, it has been suggested that the heritability of intelligence might be as high as .91 or 91 percent by the age of 65 (Posthuma et al., 2009).

One of the things that people need to understand about heritability is that estimates of heritability apply only to changes in IQ within a *group* of people, *not to the individual people themselves*. Each individual is far too different in experiences, education, and other nongenetic factors to predict exactly how a particular set of genes will interact with those factors in that one person. Only differences among people *in general* can be investigated for the influence of genes (Dickens & Flynn, 2001). Genes always interact with environmental factors, and in some cases extreme environments can modify even very heritable traits, as would happen in the case of a severely malnourished child's growth pattern. Enrichment, on the other hand, could have improved outcomes. Some observations suggest IQ scores are steadily increasing over time, from generation to generation, in modernized countries, a phenomena called the *Flynn effect* (Flynn, 2009).

7.1 7.2 7.3 7.4 7.5 7.6 7.7 7.8 7.9 7.10 7.11

THE BELL CURVE AND MISINTERPRETATION OF STATISTICS One of the other factors that has been examined for possible heritable differences in performance on IQ tests is the concept of race. (The term *race* is used in most of these investigations as a way to group people with common skin colors or facial features, and one should always be mindful of how suspect that kind of classification is. Cultural background, educational experiences, and socioeconomic factors typically have far more to do with similarities in group performances than does the color of one's skin.) In 1994, Herrnstein and Murray published the controversial book *The Bell Curve*, in which they cite large numbers of statistical studies (never published in scientific journals prior to the book) that led them to make the claim that IQ is largely inherited. These authors go further by also implying that people from lower economic levels are poor because they are unintelligent.

In their book, Herrnstein and Murray made several statistical errors and ignored the effects of environment and culture. First, they assumed that IQ tests actually do measure intelligence. As discussed earlier, IQ tests are not free of cultural or socioeconomic bias. Furthermore, as the video *In the Real World: Intelligence Tests and Stereotypes* explains, just being aware of negative stereotypes can result in an individual scoring poorly on intelligence tests, a response called **stereotype threat** (Steele & Aronson, 1995). So all they really found was a correlation between race and *IQ*, not race and *intelligence*. Second, they assumed that intelligence itself is very heavily influenced by genetics, with a heritability factor of about .80. The current estimate of the heritability of intelligence is about .50 (Plomin & DeFries, 1998).

Watch the **Video**, *In the Real World: Intelligence Tests and Stereotypes*, at **MyPsychLab**

Herrnstein and Murray also failed to understand that heritability only applies to differences that can be found *within* a group of people as opposed to those *between* groups of people or individuals (Gould, 1981). Heritability estimates can only be made truly from a group that was exposed to a similar environment.

Although *The Bell Curve* stated that Japanese Americans are genetically superior in intelligence, the book's authors overlook the influence of cultural values. Many Japanese American parents put much time and effort into helping their children with schoolwork.

One of their findings was that Japanese Americans are at the top of the IQ ladder, a finding that they attribute to racial and genetic characteristics. They seem to ignore the cultural influence of intense focus on education and achievement by Japanese American parents (Neisser et al., 1996). Scientists (Beardsley, 1995; Kamin, 1995) have concluded that, despite the claims of *The Bell Curve*, there is no real scientific evidence for genetic differences in intelligence *between* different racial groups. A series of studies, using blood-group testing for racial grouping (different racial groups have different rates of certain blood groups, allowing a statistical estimation of ancestry), found no significant relationship between ethnicity and IQ (Neisser et al., 1996).

Explore the Concept at MyPsychLab

CONCEPT MAP

Intelligence

- **individual differences** IQ tests can be used to identify individuals who differ significantly from those of average intelligence
  - **giftedness**
    - **criteria**
      - IQ > 130 (2 SD above mean)
      - IQ > 140 are called geniuses
    - **characteristics**
      - typically grow up to be well-adjusted adults EXCEPT when "pushed" to achieve at younger and younger ages
      - extreme geniuses may experience social and behavioral adjustment issues as children
  - **intellectual disability/intellectual developmental disorder**
    - **criteria**
      - IQ < 70 (2 SD below mean)
      - adaptive skills significantly below age-appropriate level
      - onset of deficits must occur during childhood or adolescence
    - **classifications**
      - range from mild to profound, depending on severity of deficits or level of support required
    - **causal factors**
      - **environmental**
        - toxins such as lead or mercury
        - poverty
      - **biological**
        - Down syndrome
        - fetal alcohol syndrome
        - fragile X syndrome
  - **emotional intelligence**
    - awareness of and ability to manage one's own emotions, self-motivation, empathy, and social skills
    - may be related to traditional intelligence but data is still being collected
- **nature vs. nurture**
  - identical twins reared together show a correlation of .86 between their IQs
    - correlation is not 1.00, so environment also has to play a part
  - heritability estimates apply *within* groups of people, not *between* groups, not to individuals, and only in a general sense
    - current heritability estimate is about .50

PRACTICE quiz How Much Do You Remember?

ANSWERS AVAILABLE IN ANSWER KEY.

Pick the best answer.

1. Kyle, age 13, has an intellectual disability complicated by multiple physical and sensory impairments that significantly impact his skills of daily living and ability to communicate. He is unable to take care of himself in any area of life. Kyle would most likely be classified with __________ intellectual disability.

a. mild
b. moderate
c. severe
d. profound

2. Lewis Terman's study provided evidence that individuals with high IQs

a. are generally weaker and lack social skills.
b. are no better at excelling in their careers than others with average IQs.
c. show little to no signs of mental illness or adjustment problems.
d. have more problems with interpersonal relationships except for those with IQs over 180.

3. What were some of the differences between the 100 most successful men and the 100 least successful men in Terman's study?

a. The successful men had higher IQ scores and better parental upbringing.
b. The successful men had higher IQ scores and no family history of mental illness.
c. The successful men had no family history of mental illness and were more motivated in general.
d. The successful men had clearly defined goals and more motivation to achieve them.

4. In recent studies, what do some researchers argue is a more accurate means of gauging success in relationships and careers?

a. intellectual intelligence
b. emotional intelligence
c. heredity studies
d. stress surveys

5. Which of the following would be an example of a stereotype threat?

a. Joaquim, who believes IQ tests are unfair to Hispanics, something that his IQ score seems to reflect
b. Jasmine, who feels she must excel on her IQ test
c. Tiana, who believes that all testing, no matter the type, is stereotypical and biased
d. Malik, who believes that tests are equal but must excel so as not to be stereotyped by his friends